Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	07 July 2026
Team ID	xxxx
Project Name	Cosmetics Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Technical Architecture – Cosmetics Insights (Tableau)

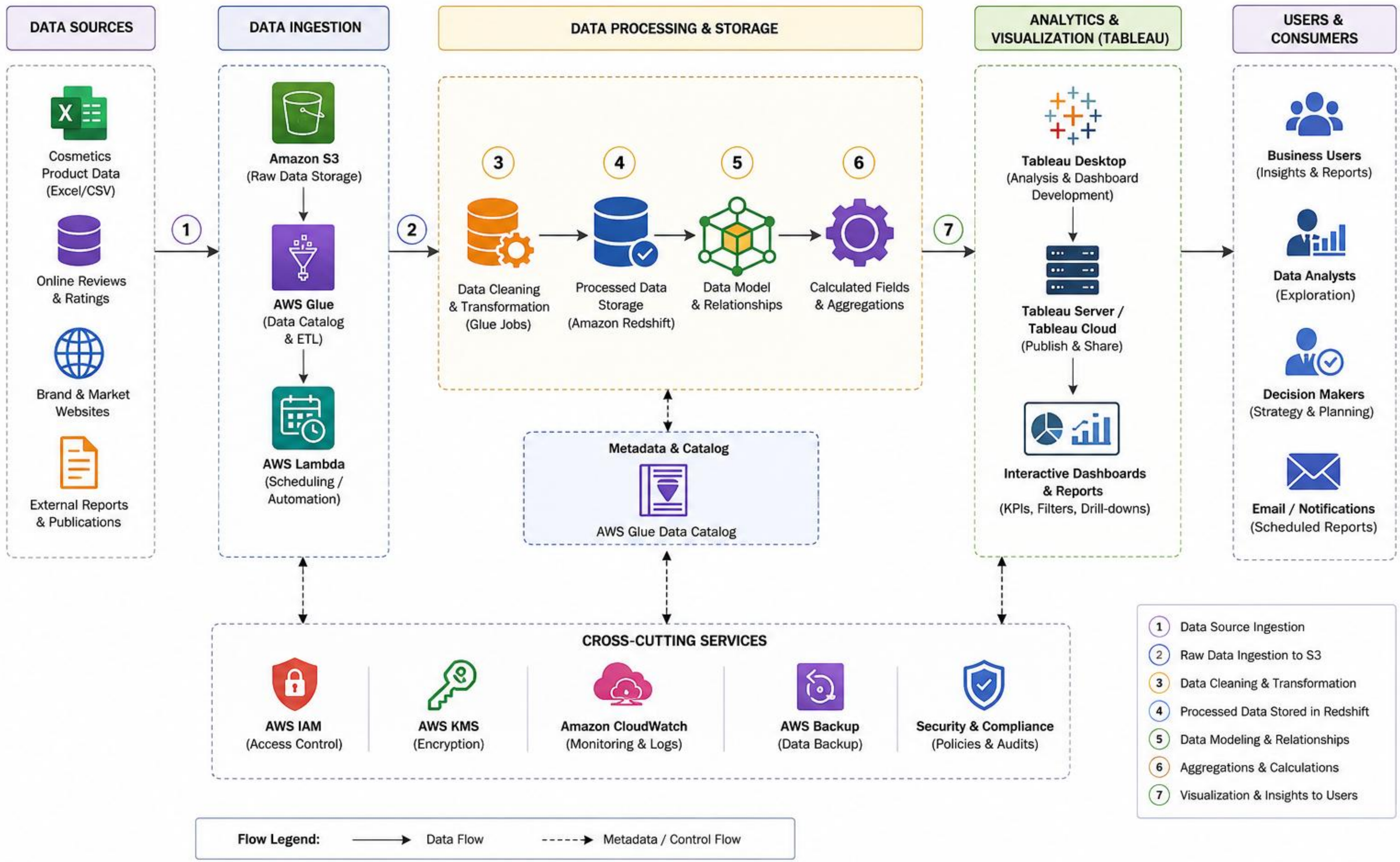

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive executive dashboards and trend explorers providing filters for consumer segments, categories, and ingredient sentiments.	Tableau Cloud / Embedded Tableau Web UI
2.	Application Logic-1	Automated ETL pipelines responsible for continuous data extraction, schema validation, normalization, and cleansing of cosmetic review datasets.	Python (Pandas, NumPy, Requests)
3.	Application Logic-2	Data orchestration layer managing workflows, dependency scheduling, and execution tasks for the data pipeline.	Apache Airflow / Prefect
4.	Application Logic-3	Data transformation and modeling layer to curate optimized, aggregate-level analytical views suited for rapid BI dashboard queries.	dbt (Data Build Tool)
5.	Database	Local metadata repository and execution logging store utilized during pipeline development and local integration phases.	PostgreSQL (v15+)
6.	Cloud Database	High-performance cloud data warehouse housing refined trend matrix tables, ingredient profiles, and historic consumer sentiment vectors.	Snowflake Data Warehouse.
7.	File Storage	Object storage repository serving as a data lake for raw JSON API payload snapshots and unstructured web-scraped historical reviews.	AWS S3 (Simple Storage Service)

8.	External API-1	Social graph and marketing analytics endpoints utilized to retrieve trending hashtags, mention volumes, and viral beauty product engagement indices.	TikTok Research API / Instagram Graph API
9.	External API-2	E-commerce API integrations used to fetch public review data, rating breakdowns, and verified consumer product descriptions.	RapidAPI / Brand-Specific Merchant Endpoints
10.	Machine Learning Model	Natural Language Processing (NLP) pipeline evaluating review sentiments, plus time series forecasting predicting next-quarter cosmetic trend cycles	HuggingFace Transformers (RoBERTa) & Meta Prophet
11.	Infrastructure (Server / Cloud)	Fully containerized pipeline infrastructure deployed via cloud container orchestrators to guarantee deterministic environments. Local Server Configuration: Docker Desktop, Minikube, 16GB RAM, Core i7. Cloud Server Configuration: AWS ECS / EKS (Elastic Kubernetes Service), Tableau Managed SaaS Environment.	AWS Cloud, Docker, Kubernetes.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Core open-source libraries enabling data ingestion, mathematical array manipulation, machine learning execution, and orchestration.	Python, Pandas, Scikit-learn, PyTorch, Apache Airflow
2.	Security Implementations	End-to-end data security ensuring data is encrypted at rest and in transit. Granular data access enforced using role-based dashboard configurations.	AES-256 Encryption, TLS 1.3, AWS IAM, Tableau Row-Level Security (RLS)
3.	Scalable Architecture	Decoupled multi-tier cloud data architecture where computing layers scale dynamically independently from storage tiers to handle multi-gigabyte text datasets.	Snowflake Virtual Warehouses, AWS Auto-scaling Groups
4.	Availability	High availability backed by automated cloud database replicas spanning multiple availability zones and managed BI cloud availability guarantees.	AWS Multi-AZ Deployment, Tableau Cloud High Availability Architecture
5.	Performance	Aggressive query acceleration using optimized workbook data hyper-extracts, data warehouse result-set caching, and pre computed analytical views.	Tableau Hyper Extracts, Snowflake Query Cache & Materialized Views

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d